

Differentiation Rules: Power, Product, Quotient, Chain

ANSWER KEY



Power, sum, and constant rules

1 Differentiate each function:

a $f(x) = 5x^4 - 3x^2 + 7$
 $= 20x^3 - 6x$

b $f(x) = \frac{1}{x^3} = -3x^{-4}$

c $f(x) = 4\sqrt{x} = \frac{2}{\sqrt{x}}$

Product and quotient rules

2 Differentiate $f(x) = (2x + 1)(x^2 - 3)$ using the product rule.

$$f'(x) = 2(x^2 - 3) + (2x + 1)(2x) = 2x^2 - 6 + 4x^2 + 2x = 6x^2 + 2x - 6.$$

3 Differentiate $f(x) = \frac{x^2 + 1}{x - 2}$ using the quotient rule.

$$f'(x) = \frac{2x(x - 2) - (x^2 + 1)(1)}{(x - 2)^2} = \frac{x^2 - 4x - 1}{(x - 2)^2}.$$

4 Differentiate $f(x) = x^2 \sin x$.

$$f'(x) = 2x \sin x + x^2 \cos x.$$

The chain rule

5 Differentiate each function using the chain rule:

a $f(x) = (3x^2 - 5)^4$
 $= 4(3x^2 - 5)^3(6x) = 24x(3x^2 - 5)^3$

b $f(x) = \sin(2x^3) = 6x^2 \cos(2x^3)$

c $f(x) = e^{-x^2} = -2xe^{-x^2}$

d $f(x) = \ln(x^2 + 1) = \frac{2x}{x^2 + 1}$

6 Differentiate $f(x) = \sqrt{4x + \cos x}$.

$$f'(x) = \frac{4 - \sin x}{2\sqrt{4x + \cos x}}.$$

Combined rules

7 Differentiate $f(x) = \frac{\sin x}{x^2}$ using the quotient rule.

$$f'(x) = \frac{x^2 \cos x - \sin x(2x)}{x^4} = \frac{x \cos x - 2 \sin x}{x^3}.$$

8 Differentiate $f(x) = e^{2x} \cos(3x)$ using the product and chain rules together.

$$f'(x) = 2e^{2x} \cos(3x) + e^{2x}(-3 \sin 3x) = e^{2x}(2 \cos 3x - 3 \sin 3x).$$

9 Differentiate $f(x) = \left(\frac{x+1}{x-1}\right)^3$ using the chain rule together with the quotient rule.

$$\text{Let } u = \frac{x+1}{x-1}, \frac{du}{dx} = \frac{(x-1) - (x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2}. \text{ So } f'(x) = 3u^2 \cdot \frac{du}{dx} = 3\left(\frac{x+1}{x-1}\right)^2 \cdot \frac{-2}{(x-1)^2} = \frac{-6(x+1)^2}{(x-1)^4}.$$

Tangent lines using derivatives

- 10** Find the equation of the tangent line to $f(x) = x^2 - 3x + 1$ at $x = 2$.
 $f(2) = 4 - 6 + 1 = -1$; $f'(x) = 2x - 3$, $f'(2) = 1$. Tangent: $y - (-1) = 1(x - 2)$, i.e. $y = x - 3$.
- 11** Find the point(s) on the curve $y = x^3 - 3x$ where the tangent line is horizontal.
 $y' = 3x^2 - 3 = 0$ gives $x = \pm 1$. Points: $(1, -2)$ and $(-1, 2)$.

Higher-order derivatives

- 12** Find $f''(x)$ for $f(x) = x^4 - 6x^2 + 2x$.
 $f'(x) = 4x^3 - 12x + 2$; $f''(x) = 12x^2 - 12$.
- 13** Find $f''(x)$ for $f(x) = \sin(2x)$.
 $f'(x) = 2\cos(2x)$; $f''(x) = -4\sin(2x)$.

Normal lines

- 14** Find the equation of the line normal (perpendicular) to $f(x) = x^2$ at $x = 2$.
 $f(2) = 4$; $f'(x) = 2x$, $f'(2) = 4$, so the tangent slope is 4 and the normal slope is $-\frac{1}{4}$. Normal line:
 $y - 4 = -\frac{1}{4}(x - 2)$, i.e. $y = -\frac{1}{4}x + \frac{9}{2}$.

Visualizing a function and its tangent line

- 15** The function $f(x) = x^2 - 3x + 1$ and its tangent line at $x = 2$ (found earlier, $y = x - 3$) are graphed together. Use the graph to confirm the tangent line touches the curve at exactly the point $(2, -1)$.
At $x = 2$: $f(2) = 4 - 6 + 1 = -1$ and the line gives $y = 2 - 3 = -1$ — both equal -1 , confirming the tangent line passes through $(2, -1)$, matching the point of tangency shown on the graph.

Differentiating inverse trig and more transcendental functions

- 16** Differentiate each function:
- a** $f(x) = \tan^{-1}(3x) = \frac{3}{1+9x^2}$
 - b** $f(x) = \sin^{-1}(x^2) = \frac{2x}{\sqrt{1-x^4}}$
 - c** $f(x) = x \ln x = \ln x + 1$ (product rule)

Implicit differentiation (a preview)

- 17 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$ by differentiating both sides with respect to x (treating y as a function of x).
 $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}.$
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Related applications of the derivative rules

- 18 A particle's position is $s(t) = t^3 - 6t^2 + 9t$. Find its velocity function $v(t) = s'(t)$ and determine when the particle is at rest ($v(t) = 0$).
 $v(t) = 3t^2 - 12t + 9 = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3)$. At rest when $t = 1$ or $t = 3$.